

ISE549 | Week 2 | Creativity, Innovation Types & Open Innovation

創意、創新類型與開放式創新

Lecture: Creativity and Innovation, The “WOW” effect, Innovation processes and systems, Types of Innovations

1. Core Themes | 核心主題

1. Creativity vs. Innovation | 創意與創新

Creativity produces novel possibilities; innovation turns selected ideas into accepted and sustainable value.

創意產生新奇可能性；創新則把選定想法轉化為被接受且可持續的價值。

2. Innovation as a process | 創新是一個過程

Innovation moves from opportunity and problem identification to testing, commercialisation and diffusion.

創新由機會與問題識別開始，經測試、商業化及擴散而完成。

3. Innovation types matter | 創新類型影響策略

Firms must identify whether change affects components, architecture, or both, because each type requires different capabilities and carries different risks.

企業需判斷改變發生在組件、架構或兩者，因為不同類型需要不同能力及承擔不同風險。

4. Disruption and strategic traps | 顛覆與策略陷阱

Past success can become an evolutionary trap when firms ignore low-end entrants, changing technologies or new market needs.

當企業忽略低端新進者、技術改變或新市場需求時，過去成功可能變成演化陷阱。

5. Open innovation and ecosystems | 開放式創新與生態系統

Firms create more value by combining internal capabilities with external knowledge, partners, customers and developers—while protecting core assets.

企業透過結合內部能力與外部知識、夥伴、顧客及開發者創造更多價值，同時必須保護核心資產。

2. Key Concepts | 重要概念

2.1 Creativity | 創意

- **English definition:** The generation of new, unusual, surprising or conflicting ideas.
- **中文解釋:** 產生新穎、不尋常、令人驚喜或具衝突性的想法。
- **Example | 例子:** A restaurant serving food on a book-like “plate” creates an unexpected combination.
餐廳以看似書本的「餐盤」盛載食物，產生違反預期的組合。
- **Exam point | 考試重點:** Creativity creates possibilities, but it is not automatically innovation.
創意產生可能性，但不會自動成為創新。

2.2 Innovation | 創新

- **English definition:** The application and commercialisation of ideas, knowledge or technologies to create accepted value.
- **中文解釋:** 把想法、知識或技術應用及商業化，以創造被接受的價值。
- **Example | 例子:** A smart water bottle becomes innovation only after testing, pricing, customer adoption and improvement.
智能水瓶只有在經過測試、定價、顧客採用及持續改善後，才成為創新。
- **Exam point | 考試重點:** Must be implementable, reliable, scalable and sustainable.
必須可實施、可靠、可擴展及可持續

2.3 Incremental / Modular / Architectural / Radical Innovation

漸進式／模組式／架構式／激進式創新

- **English definition:** Innovation types classified by the degree of change in components and the links between them.
- **中文解釋:** 根據組件本身及組件之間連結改變的程度，區分不同創新類型。
- **Example | 例子:** Replacing a traditional battery with a solar-powered module while keeping the same product system is modular innovation.
以太陽能模組取代傳統電池，但維持原有產品系統，屬於模組式創新。
- **Exam point | 考試重點:**
 - Incremental = small improvement | 小幅改善
 - Modular = components change | 組件改變
 - Architectural = links change | 組件連結改變
 - Radical = components + architecture change | 組件與架構均大幅改變

2.4 Disruptive Innovation | 顛覆性創新

- **English definition:** A simpler or more accessible offering that begins in low-end or overlooked markets, improves over time, and challenges incumbents.
- **中文解釋:** 新進者以較簡單、便宜或容易接觸的方案切入低端或被忽略市場，逐步改善後挑戰既有企業。

- **Example | 例子：** Low-cost airlines provide basic, affordable flights for customers who do not need full-service travel.
低成本航空為不需要全套服務的顧客提供基本且便宜的航班。
- **Exam point | 考試重點：** Not every new technology is disruptive; it must show the low-end/new-market entry and improvement pattern.
並非所有新科技都是顛覆性創新；必須呈現由低端／新市場切入並持續改善的模式。

2.5 Open Innovation | 開放式創新

- **English definition:** The strategic use of internal and external knowledge flows, technologies and partners to create and commercialise innovation.
- **中文解釋：** 策略性地運用內部及外部知識流、技術及合作夥伴，以創造及商業化創新。
- **Example | 例子：** A medical-device firm collaborates with a university, hospital, manufacturer and patients.
醫療器械企業與大學、醫院、製造商及病人共同合作。
- **Exam point | 考試重點：** Open innovation improves access to knowledge and speed, but creates IP, coordination and control risks.
開放式創新可取得更多知識並加快開發，但會帶來知識產權、協調及控制權風險。

2.6 Innovation Ecosystem | 創新生態系統

- **English definition:** A network of interdependent actors that collaborate to create, deliver and capture innovation value.
 - **中文解釋：** 由相互依賴的參與者組成、共同創造、交付及取得創新價值的網絡。
 - **Example | 例子：** Customers, suppliers, universities, investors, governments and developers jointly support innovation.
顧客、供應商、大學、投資者、政府及開發者共同支援創新。
 - **Exam point | 考試重點：** A good technology alone is insufficient; firms also need adoption, funding, capabilities, channels and complementary partners.
單有好技術並不足夠；企業還需要採用者、資金、能力、渠道及互補夥伴。
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3. Important Comparisons | 重要比較

Comparison 比較	Key Difference 主要分別	Exam Use 考試用途
Creativity vs. Innovation	Creativity generates ideas; innovation implements ideas to create accepted value. 創意產生想法；創新把想法實施並創造被接受的價值。	Explain why novelty or a WOW effect alone is insufficient. 說明為何新奇或 WOW effect 本身不足夠。
Incremental vs. Radical	Incremental improves an existing system; radical changes technologies, knowledge or architecture significantly. 漸進式改善現有系統；激進式大幅改變技術、知識或架構。	Compare risk, uncertainty, customer acceptance and capability needs. 比較風險、不確定性、顧客接受度及能力要求。
Creative Destruction vs. Disruptive Innovation	Creative destruction reshapes the economy broadly; disruption describes entrants challenging incumbents through low-end/new-market entry. 創造性破壞重組整體經濟；顛覆性創新描述新進者由低端／新市場挑戰既有企業。	Avoid treating the two theories as identical. 避免把兩種理論混為一談。
Closed vs. Open Innovation	Closed relies mainly on internal R&D; open combines internal and external knowledge flows. 封閉式主要依賴內部研發；開放式結合內外部知識流。	Evaluate control/IP protection against speed, knowledge access and ecosystem benefits. 評估控制及 IP 保護，相對於速度、知識取得及生態系統好處。
Innovation Process vs. Incremental Innovation	A process can develop step by step even if the final innovation is radical. 即使最終是激進式創新，發展過程仍可逐步進行。	Useful for correcting a common conceptual error. 用於糾正常見概念錯誤。

4. Key Case | 核心案例：NVIDIA CUDA

Case summary | 案例摘要

NVIDIA uses CUDA tools, SDKs, libraries, education, documentation and developer communities to encourage external developers to build applications on its platform.

NVIDIA 利用 CUDA 工具、SDK、函式庫、教育資源、文件及開發者社群，吸引外部開發者在其平台建立應用程式。⁸

Strategic logic | 策略邏輯

More tools and learning resources → More developers

→ More CUDA-optimised applications → More GPU demand

→ Higher switching costs

更多工具與學習資源 → 更多開發者 → 更多 CUDA 優化應用 → 更高 GPU 需求

→ 更高轉換成本

Exam use | 考試用途

- **Open innovation:** external developers co-create value.
開放式創新：外部開發者共同創造價值。
 - **Innovation ecosystem:** tools, developers, hardware and users reinforce each other.
創新生態系統：工具、開發者、硬件及用戶互相強化。
 - **Selective openness:** NVIDIA opens selected resources but protects proprietary core technologies.
選擇性開放：NVIDIA 開放部分資源，但保護專有核心技術。
 - **User lock-in:** developers face high switching costs after investing in CUDA skills, code and workflows.
用戶鎖定：開發者投入 CUDA 技能、程式碼及工作流程後，轉換成本會提高。
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5. Exam Answer Framework | 考試答題框架

Define → Explain → Apply → Critique → Conclude

定義 → 解釋 → 應用 → 評估 → 結論

1. **Define | 定義**

State the concept precisely.

清楚寫出概念的準確定義。

2. **Explain | 解釋**

Explain its mechanism, key features or conditions.

解釋其運作機制、核心特徵或成立條件。

3. **Apply | 應用**

Apply the concept to the case evidence.

把理論套用到案例事實。

4. **Critique | 評估**

Discuss limitations, risks, trade-offs or conditions for success.

討論限制、風險、取捨或成功條件。

5. **Conclude | 結論**

Give a balanced judgment.

作出平衡的判斷。

Useful exam sentence | 可直接使用的考試句：

This case demonstrates ___ because ___. However, its success depends on ___, and the firm must manage ___.

此案例展示了 ___, 因為 ___. 然而，其成功取決於 ___, 而企業必須管理 ___。

6. One-Minute Revision | 一分鐘溫習版

1. Creativity = novel possibilities; Innovation = accepted value.
創意＝新奇可能性；創新＝被接受的價值。
2. WOW attracts attention, but value requires implementation and adoption.
WOW 吸引注意，但價值需要實施及採用。
3. Innovation process: idea → test → refine → commercialise → diffuse.
創新過程：想法 → 測試 → 改善 → 商業化 → 擴散。
4. Do not confuse a step-by-step process with incremental innovation.
不要把逐步發展過程誤當作漸進式創新。
5. Incremental = small improvement; Radical = major technological/architectural change.
漸進式＝小幅改善；激進式＝重大技術／架構改變。
6. Disruption starts from overlooked markets and moves upward.
顛覆通常由被忽略市場切入，再逐步上移。
7. Past success can create an evolutionary trap.
過去成功可能形成演化陷阱。
8. Open innovation combines internal and external knowledge.
開放式創新結合內部及外部知識。
9. Open the boundary, but protect IP and core capabilities.
開放邊界，但要保護知識產權及核心能力。
10. NVIDIA CUDA shows ecosystem growth, selective openness and user lock-in.
NVIDIA CUDA 展示生態系統成長、選擇性開放及用戶鎖定。

Memory phrase | 記憶句：

Generate ideas, create value, collaborate wisely, and protect the core.

產生想法、創造價值、聰明合作、保護核心。